Phase 3: Design Rationale & Self-Audit

This document explains the redesign of **Case B**, the Gapminder bubble chart, as a self-contained rationale. It is written so that a reader who has not seen the implementation code can still understand what was changed, why it was changed, and what trade-offs remain. [1] [2] [3] [4]

Section A: Design Rationale

I chose **Case B** because it offered the clearest opportunity to repair bias through interaction design rather than only through cosmetic restyling. In the original Gapminder view, animation, logarithmic scaling, and large bubble salience work together to guide interpretation very strongly, so a redesign could meaningfully reduce bias while still preserving the same core variables and overall topic. [1] [2] [3]

My design process followed the **Visualization Design Process** framework from lecture. In **Stage 1**, I treated the assignment as a problem-framing task rather than a style exercise. Phase 1 showed that the original chart's strongest issues were not simply aesthetic: autoplay encouraged a smooth progress narrative, the default x-axis hid a consequential log-scale choice, and fluid movement overstated continuity in the underlying historical record. [1] [2] [3] This led to a central redesign goal: keep the same topic and variables, but replace **cinematic persuasion** with **deliberate comparison**.

In **Stage 2**, I translated those diagnosed biases into concrete representational choices. The first key decision was to replace autoplay as the default with **small multiples for fixed comparison years**. This directly responds to lecture material on multiple-object tracking limits and on the trade-off between animation and small multiples for temporal comparison. [3] A second decision was to expose the income axis as a visible **log/linear toggle**. In the original interface, logarithmic spacing is analytically plausible but rhetorically hidden. By making scale switchable, the redesign turns a concealed modeling assumption into something the viewer can inspect. A third decision was to add a visible **method note** stating that the redesign remains faithful to Gapminder's core variables while using a five-year sampled teaching dataset. No synthetic

observations were added, but the redesign surfaces the sampling structure so the historical record does not appear smoother or more continuous than it really is. [4]

In **Stage 3**, I implemented and refined the interface as an evidence-focused reading environment. The page uses an **archival atlas** visual language with restrained typography, pale drafting textures, and a left evidence rail. This choice was intentional: a glossy dashboard aesthetic would have recreated the same presentation energy that the redesign was trying to reduce. During testing, I also compressed the hero section so that the analytical content appears earlier above the fold, because the first screen needed to communicate comparison-first reading, not decorative atmosphere.

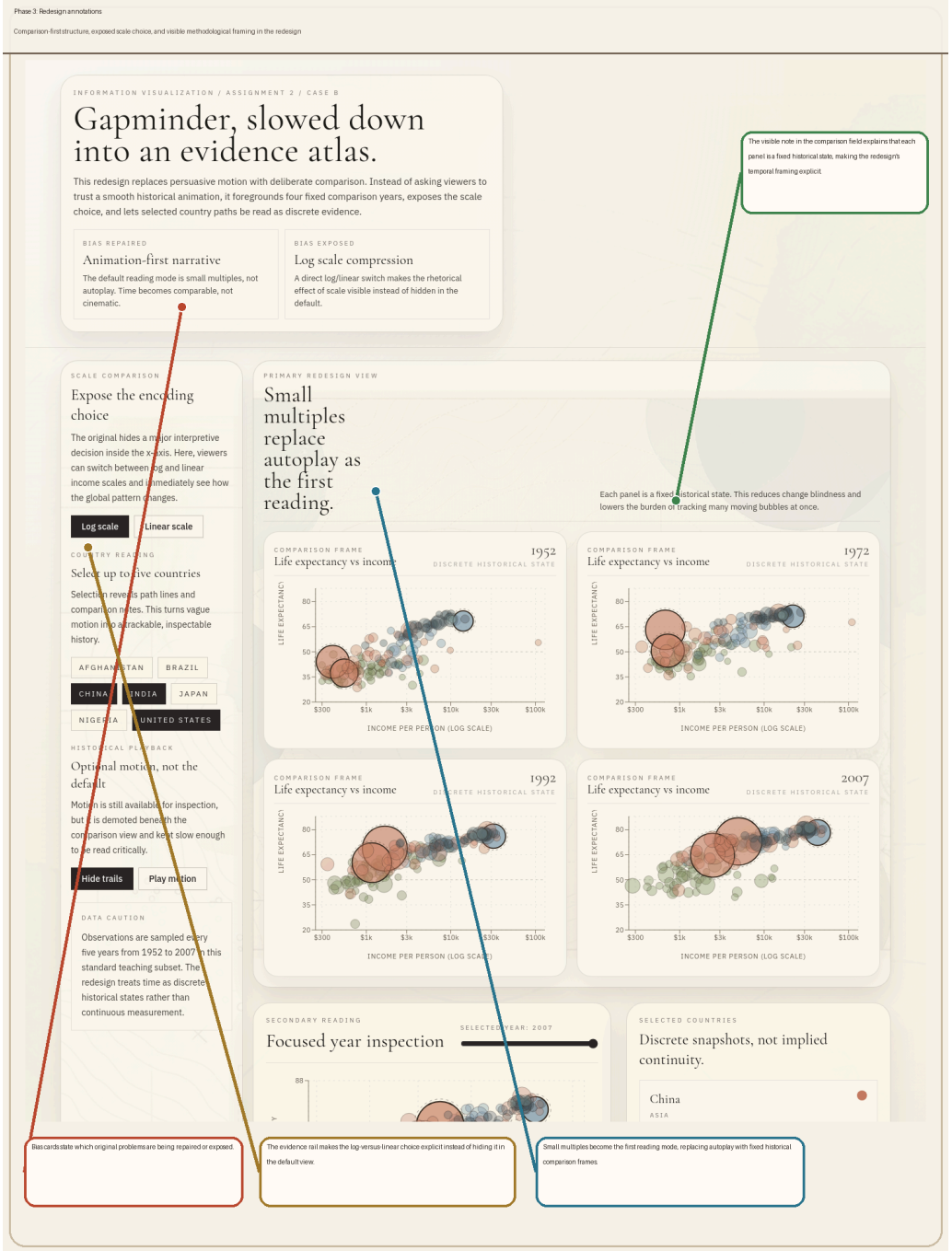

Figure 1. Annotated redesign screenshot showing explicit bias-repair framing, the exposed scale choice, the comparison-first structure, and the visible methodological note.

Section B: Before/After Comparison

The comparison below shows the same conceptual case before and after redesign. In the **original** view, the interface invites viewers to press play and watch a large field of moving bubbles, with the huge year label reinforcing a cinematic sense of progression. The logarithmic income axis is present, but its interpretive consequences are not made explicit. In the **redesign**, the opening state is four fixed comparison years rather than autoplay, so time is first encountered as evidence that can be compared side by side. The income-axis decision is exposed through a direct **log/linear switch**, and a visible methodological note warns that the teaching dataset is sampled in five-year snapshots. These changes address the main biases diagnosed in Phase 1: **animation-first narrative**, **multiple-object tracking overload**, **hidden log-scale rhetoric**, and **overstated continuity of the historical record**. [2] [3] [4]

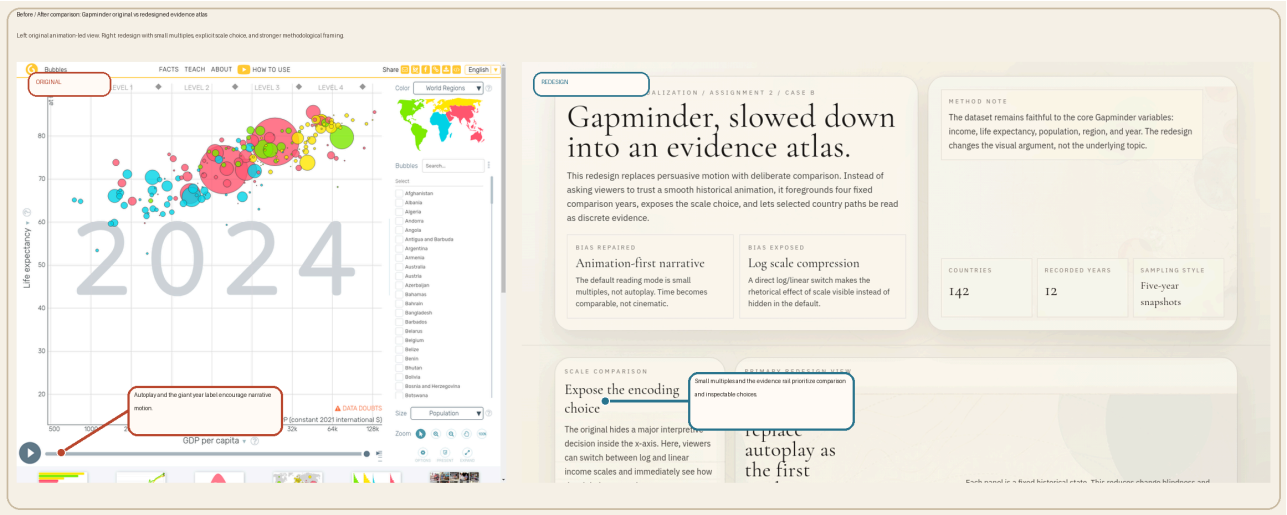

Figure 2. Side-by-side comparison of the original Gapminder view and the redesigned evidence-atlas view, with annotations highlighting the most important changes.

Section C: Self-Audit

The redesign improves interpretability, but it does not eliminate every bias. First, **bubble area** still represents population, which means very large countries remain visually prominent. This is not accidental; the assignment requires the same core

variables to be preserved, and population is one of the core Gapminder encodings. I reduced the problem by shifting the interface toward comparison and selective focus, but the underlying salience of large population bubbles remains. Second, even after exposing the **log/linear switch**, some viewers may still prefer the log view without critically reflecting on what it changes. Making an assumption visible is better than hiding it, but visibility alone does not guarantee critical reading. Third, the redesign still uses a scatterplot-based framing, so overlap and density can remain challenging when many countries occupy similar positions. [2] [3]

If I had more time, the next improvement I would make would be a stronger **uncertainty layer** for historical sampling and data quality. At present, the redesign communicates five-year snapshot sampling through text and framing, but it does not yet visually grade country-years by confidence, interpolation, or incompleteness. Implementing that carefully would take more time because it requires additional metadata and a design that communicates uncertainty without overwhelming the main comparison task. I chose not to force that feature into the current assignment build because a weakly justified uncertainty overlay could easily create a new layer of confusion rather than a real improvement.

One alternative path I considered was a **story-driven scrollytelling redesign** that would walk the viewer through selected countries or regions with curated narrative checkpoints. That approach could have produced a stronger authored explanation and clearer takeaways for novice audiences. However, it would also have traded away the redesign's main goal, because it would replace one persuasive narrative structure with another rather than making the chart more inspectable and comparison-oriented.

References

1. **Assignment 2 brief.** Uploaded file: `Assignment2(1).pdf`.
2. **Phase 1 diagnosis.** Project file: `phase1_diagnosis.md`.
3. **CS5346 Lecture 10.** Uploaded file: `CS5346-2026_Lecture10.pdf`.
4. **Plotly Gapminder dataset.** https://github.com/plotly/datasets/blob/master/gapminder_with_codes.csv